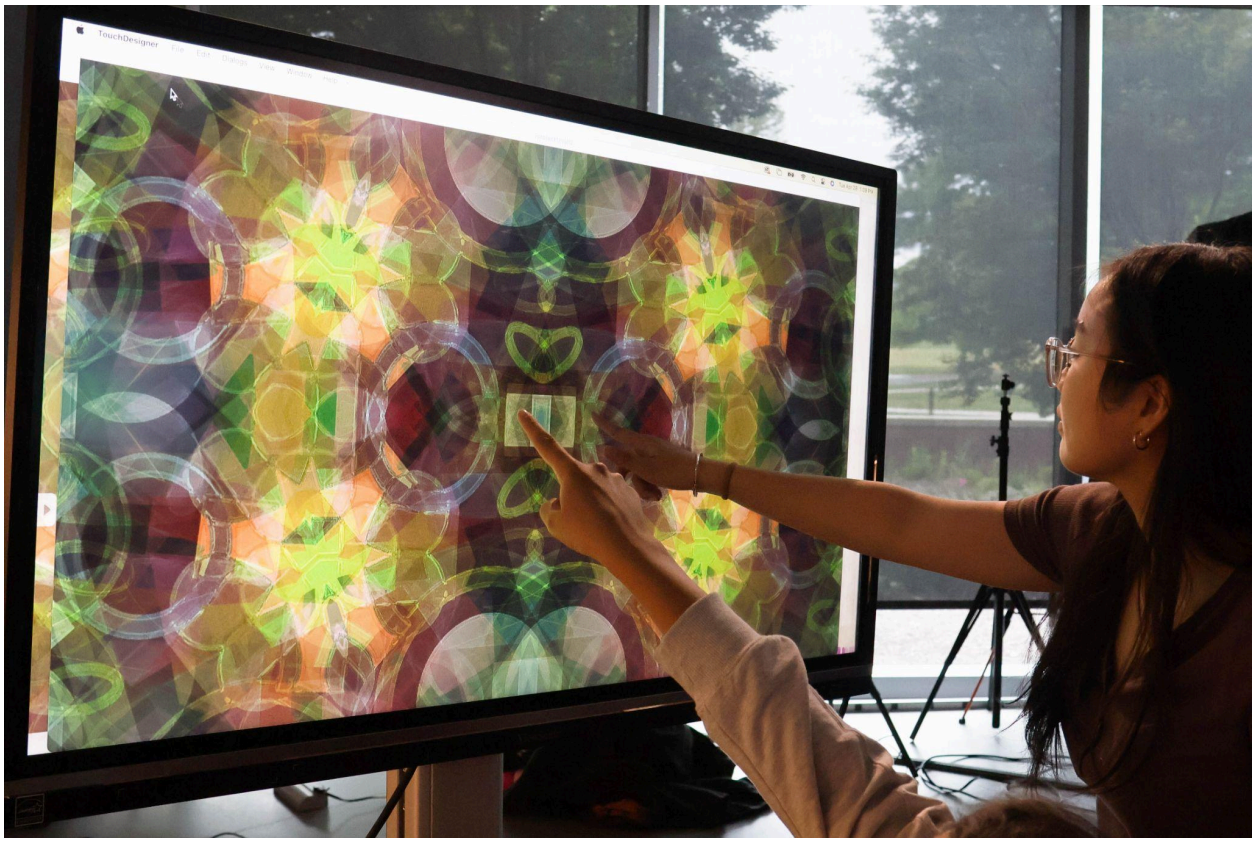
Aroma Synesthesia

By: Adrianna Hwang and Hannah Chen



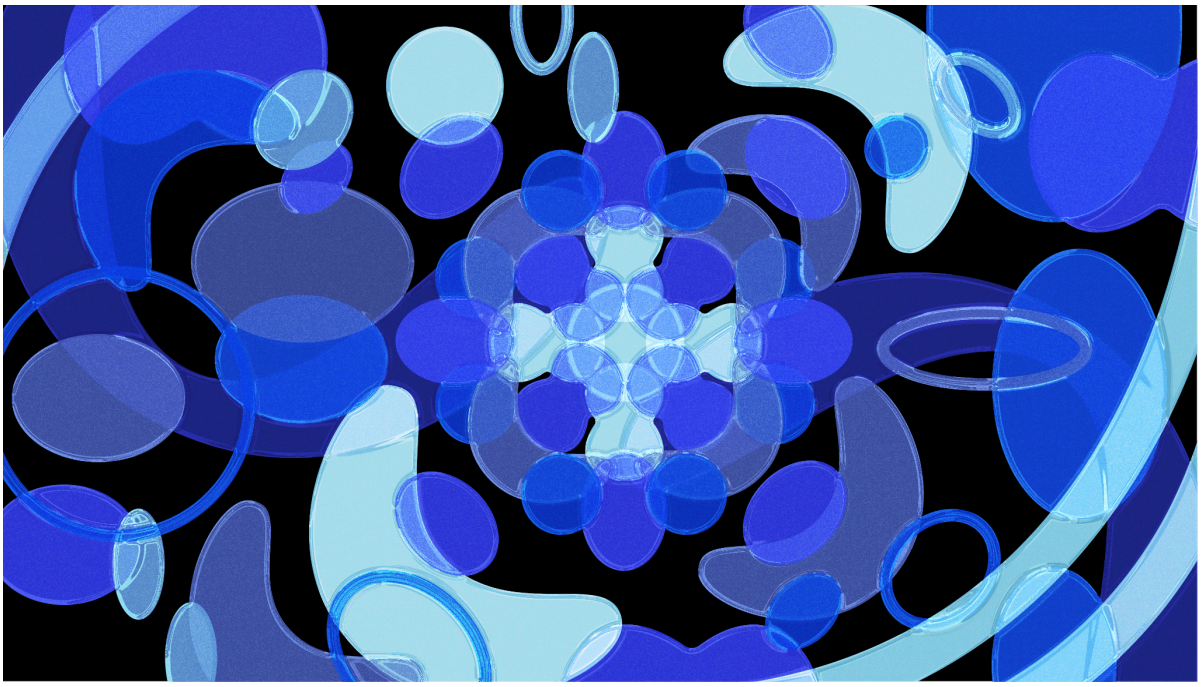
“Aroma Synesthesia” is an interactive brainwave based art piece that outputs a unique kaleidoscope pattern based on what the EEG wearer smells. The key concept of this piece is to generate art through the underutilized sense of smell and attempt to visualize the affects of fragrance. Considering the gamma, beta, alpha, and theta brainwaves that the EEG headset tracked, we created 4 representative visuals for each respective wave state and then layered them. Based on the strength of a certain reaction, the representative image’s opacity would become stronger. So, depending on the strength of a given wave, the EEG wearer would create a unique layering of visuals when they interacted with a scent.

Our original concept was for the user to wear an EEG headset while inhaling various scents specifically chosen to "activate" specific brainwaves. The brainwave frequencies’ data would be transmitted from the EEG to Python Libraries. We would manipulate the data in Touch Designer, using it to impact the opacity and rotation speed for each brainwaves’ corresponding pattern, and the pitch and speed of the music.

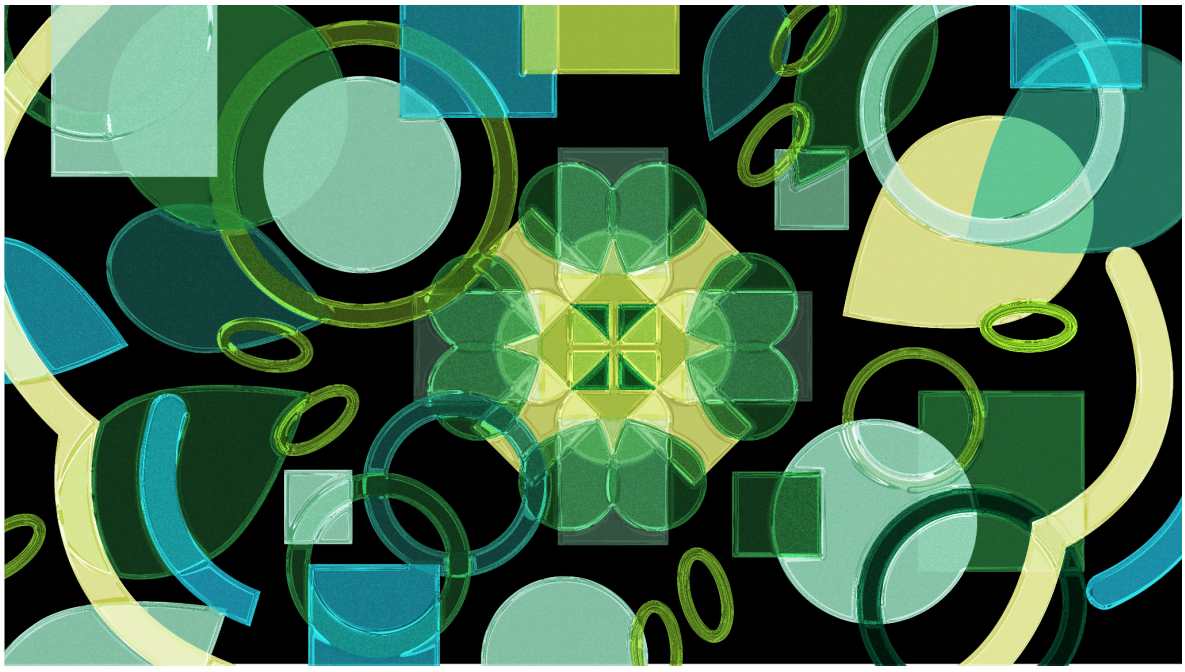
The EEG processed 4 key waves: gamma, beta, alpha, and theta. Gamma was the most excited state with the most frequent waves, while theta was the calmer state with longer waves. Based on this range, we represented gamma waves with bright reds and sharp, excited visuals while theta was represented with blue colors and softer rounded shapes. These extreme ends

would be represented by more organic, irregular shapes while the middle two were more geometric to introduce an in between. So, beta trended towards a more excited state represented with warmer tones of pink and yellow, not as bright as red and alpha towards a calmer state represented with cooler tones of green, with yellowish tints to suggest slight excitement.

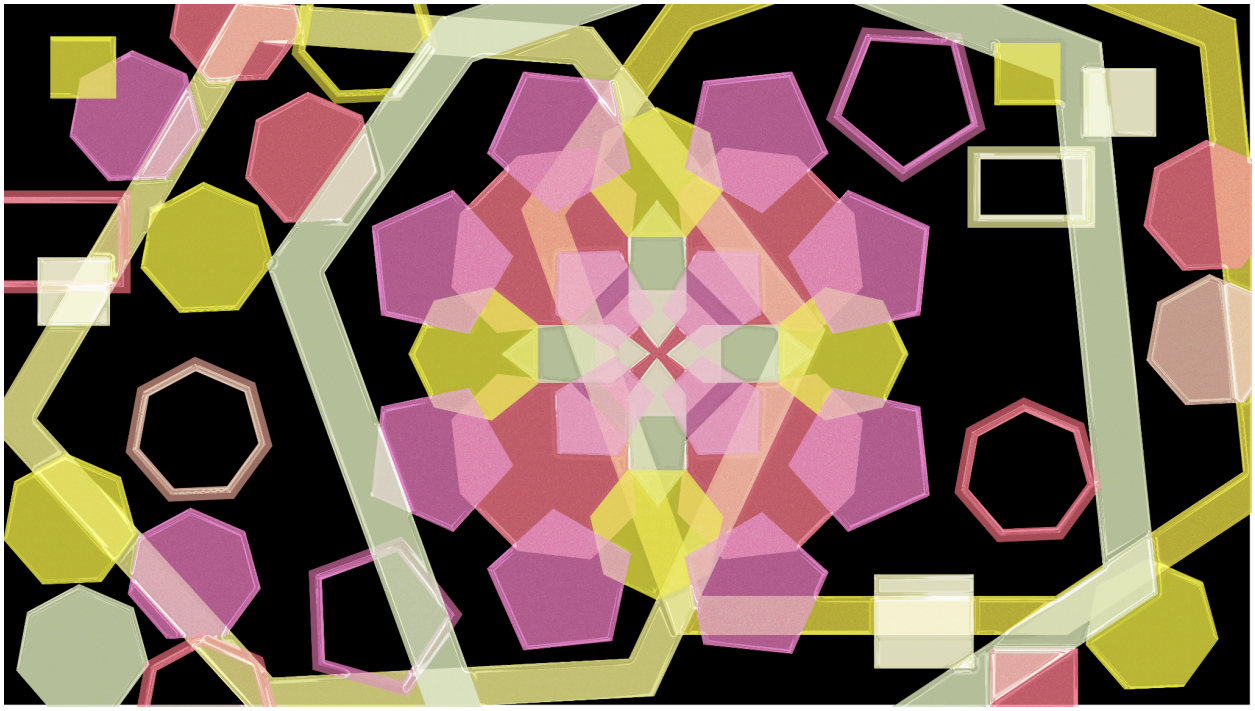
Theta wave pattern:



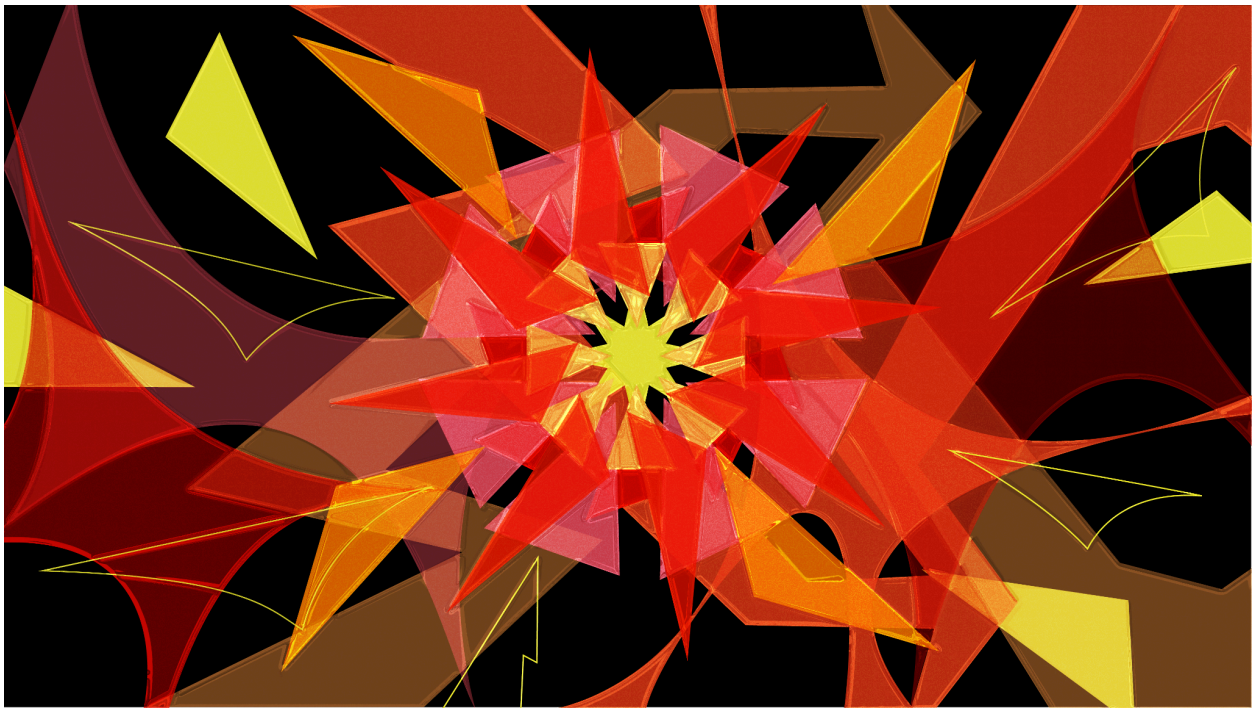
Alpha wave pattern:



Beta wave pattern:



Gamma wave pattern:

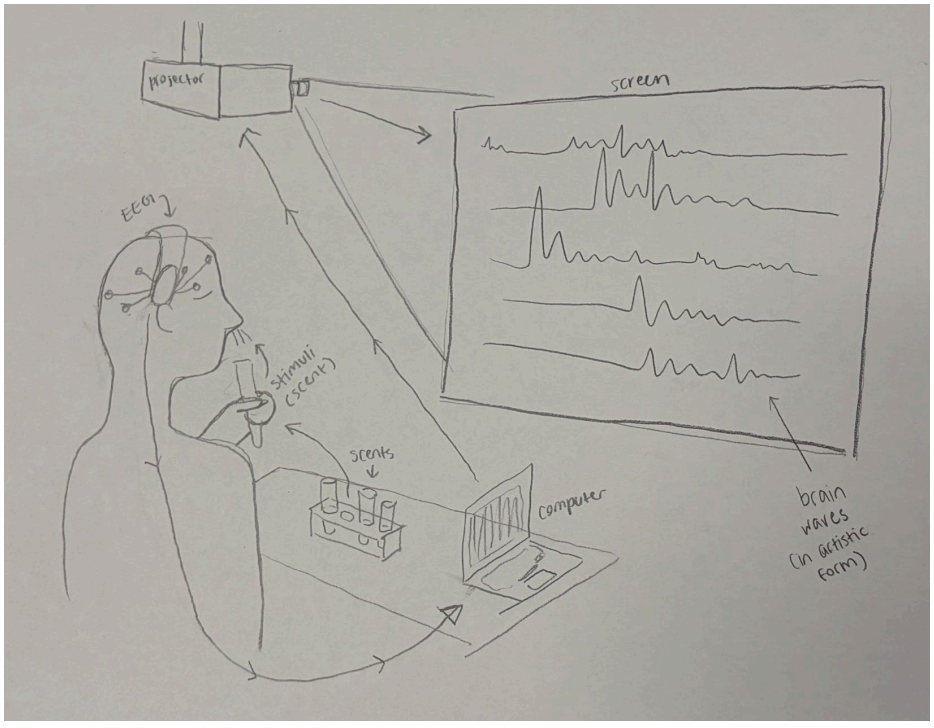


We created abstract patterns like these to more clearly represent the emotional aspect our brainwaves are associated with. From our experimentation, we observed that the brainwave frequencies were consistently at similar levels, regardless of our pool of scent stimuli. This

contrasted with our initial belief that the frequency of a singular brainwave would be drastically different based on a single stimulus. Therefore chose to layer them because it better reflected how the EEG operates. Additionally, we wanted the user to be able to see a visualization of their own unique combination of brainwave frequencies which is easier to understand, instead of data on a screen, and how these frequencies change over time and react to different stimuli.

Scent is heavily correlated with emotion and memory, being one of the primary senses people recall. The capacity for fragrance to unlock uniquely subjective emotional responses brings an interesting research question forward. Before we were able to use the EEG, we expected a cause and effect relationship between smell and brainwave response, where a specific smell, like lavender, would produce an expected response, calmer waves. However, the waves were more continuous and less differentiated than that. So, similarly to how we adapted our visuals to more accurately represent the user's mental state through layering, we needed to approach scent's role differently. Instead of a strong cause and effect smell relationship, we came to the conclusion that scent should be used more experimentally. Even when different users were just wearing the EEG, their visual results were unique simply based on their physical and mental states. If we gave a user a certain scent, we would instead lean into the subjectivity of their response to the stimuli. For example, some people may appreciate the scent of flowers and feel calm or content, but some people might hate the smell of flowers because it reminds them of bad experiences. This potential variability would provide us with insight into the relationship between the smells we chose and their experiences with said smell.

Diagram Plan



We received positive feedback, particularly with the visuals of our project. The novelty of an EEG headset and a personalized experience, based on each individual's brainwaves, also contributed to the positivity of the audience's response. However, due to the nature of how the EEG tracks and outputs data, the rotation of the kaleidoscope was choppy because the data updated every few seconds, setting the axis back to 0. So, to improve the visual immersion, smoothing out this effect will allow for a better experience.

From this project, we learned how brainwaves function: all our brainwaves are simultaneously activated and not just one at a time. From this insight, we had to adjust our concept, making it more experimental rather than a cause and effect- based project. Additionally we also learned how to use an EEG, Python Libraries, how to connect the two technologies together, and how to implement or manipulate the data in Touch Designer. Overall, we were able to achieve our goal of creating an interactive, EEG- based art project.

Github Repository: https://github.com/hchen114/IMD-Lab-SmellEEG